

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1039+350

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

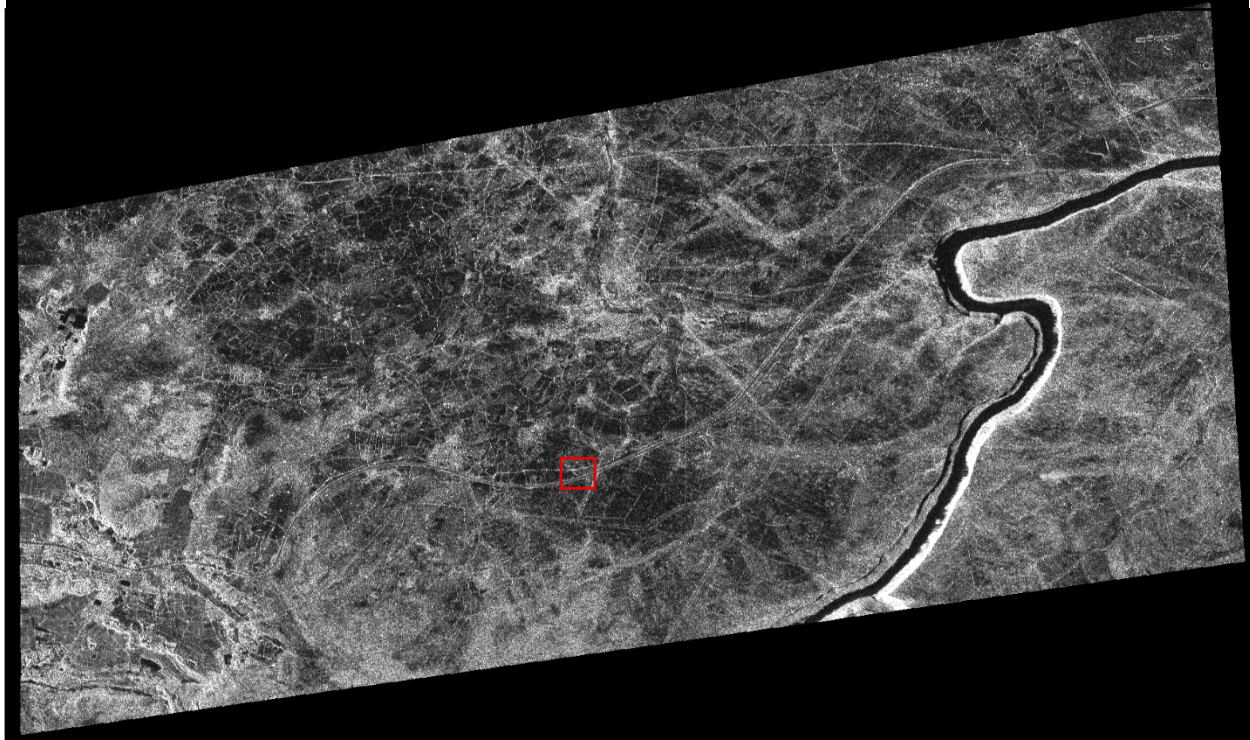
Flight Direction - DESCENDING

Polarization – VV+VH

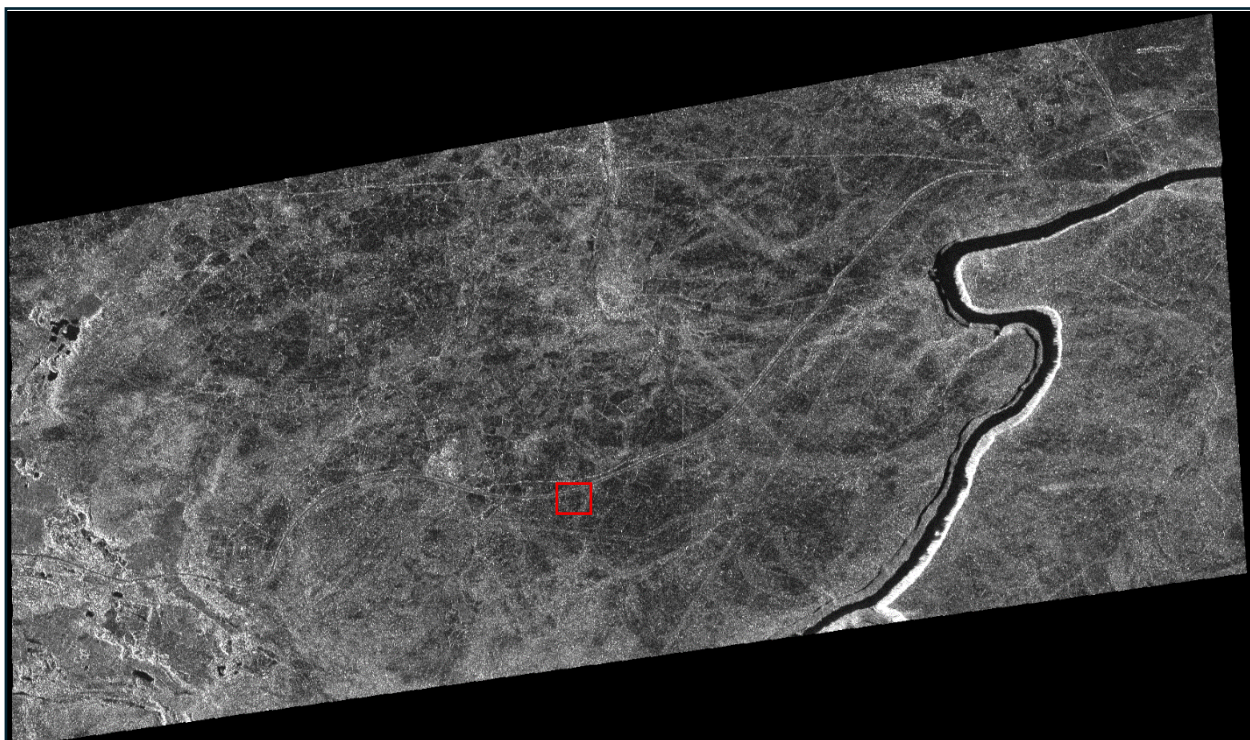
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

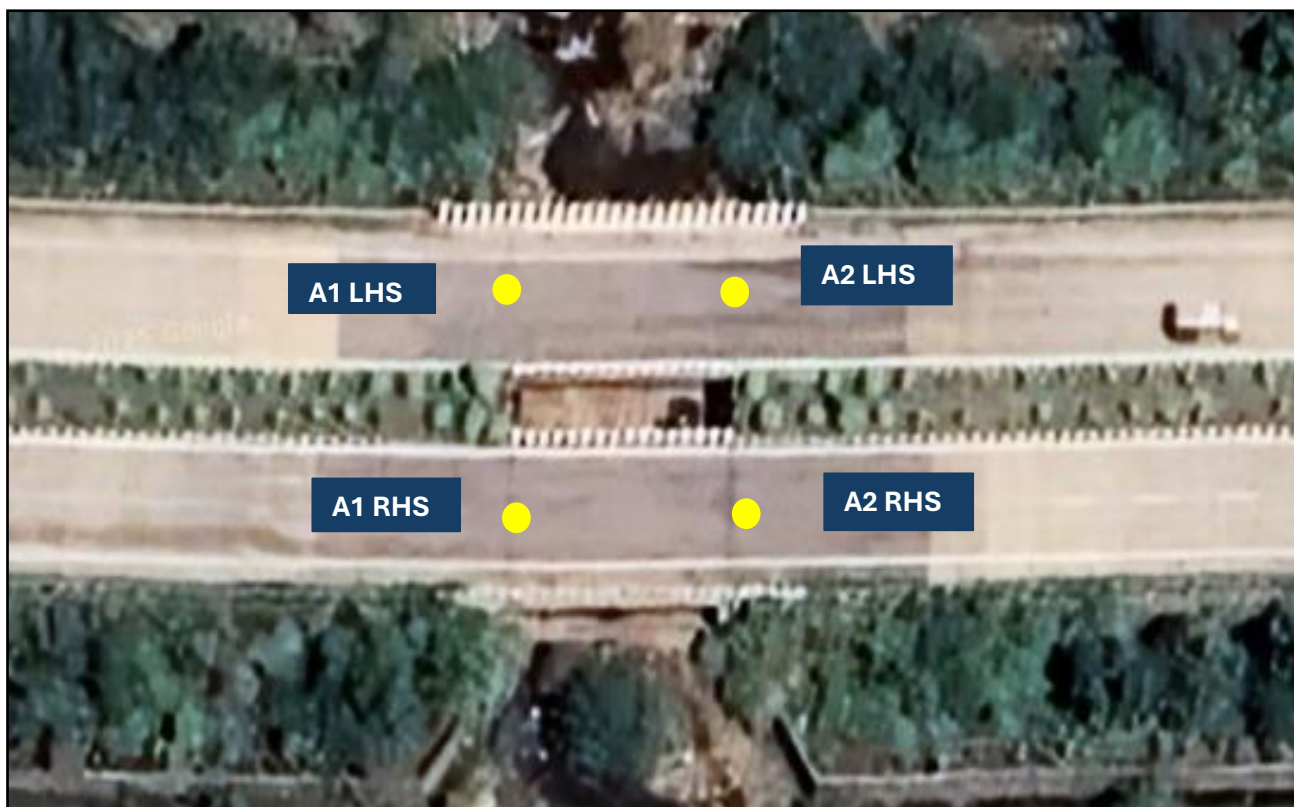


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

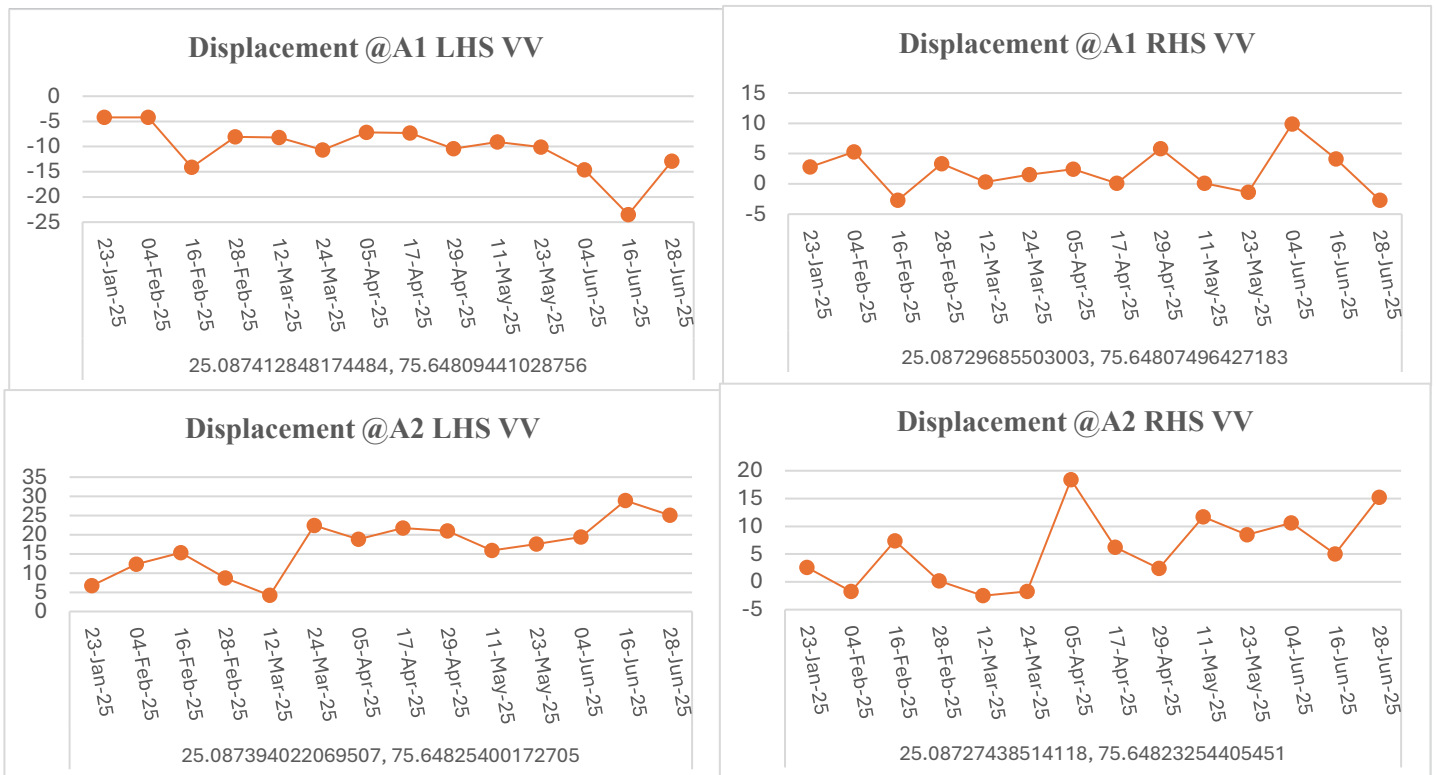
MNB – CH-1039+350

Length – 26.94M

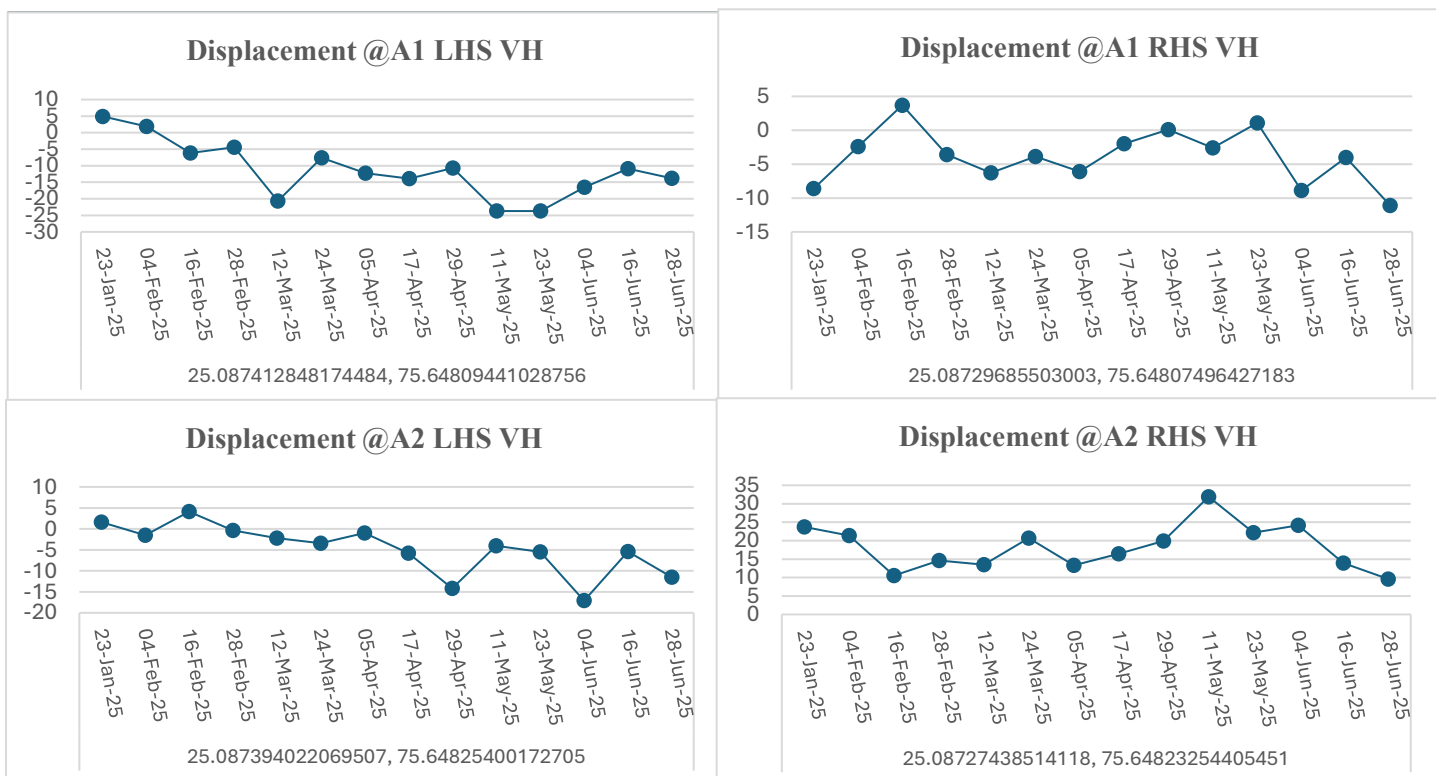


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement from 23 Jan 2025 to 28 June 2025



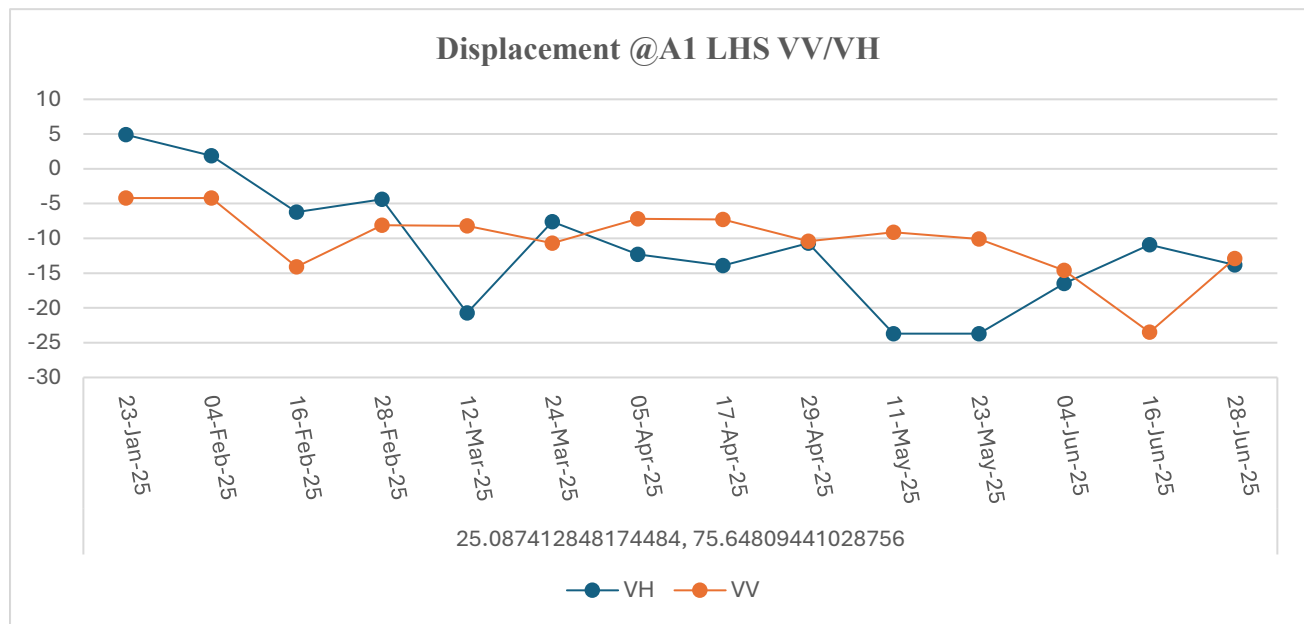
3.2 VH Displacement from 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.087412848174484, 75.64809441028756)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the A1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with a strong negative dominance, ranging roughly between +4 mm and –26 mm.

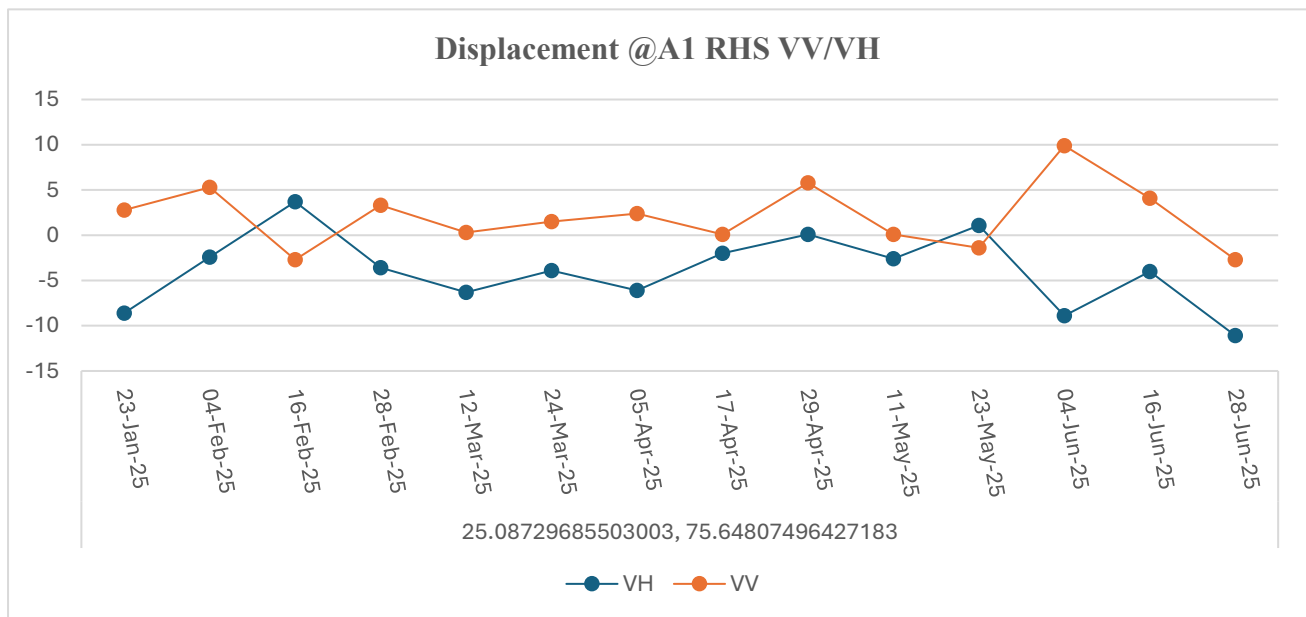
Critical Events & Anomalies:

- **February 2025:** VH drops sharply to –7 mm, while VV decreases to –14 mm, marking a distinct coupled settlement event affecting both horizontal and vertical components.
- **May 2025:** VH improves marginally to –16 mm, while VV declines further to –25 mm, indicating renewed vertical settlement despite lateral relief.
- **June 2025:** VH recovers slightly toward –10 mm, while VV improves to –12 mm, suggesting partial post-event adjustment but no full stabilization by the end of the monitoring window.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.08729685503003, 75.64807496427183)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits predominantly negative values with intermittent recoveries, ranging roughly between +3 mm and –12 mm.
- **Vertical displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –3 mm to +10 mm.

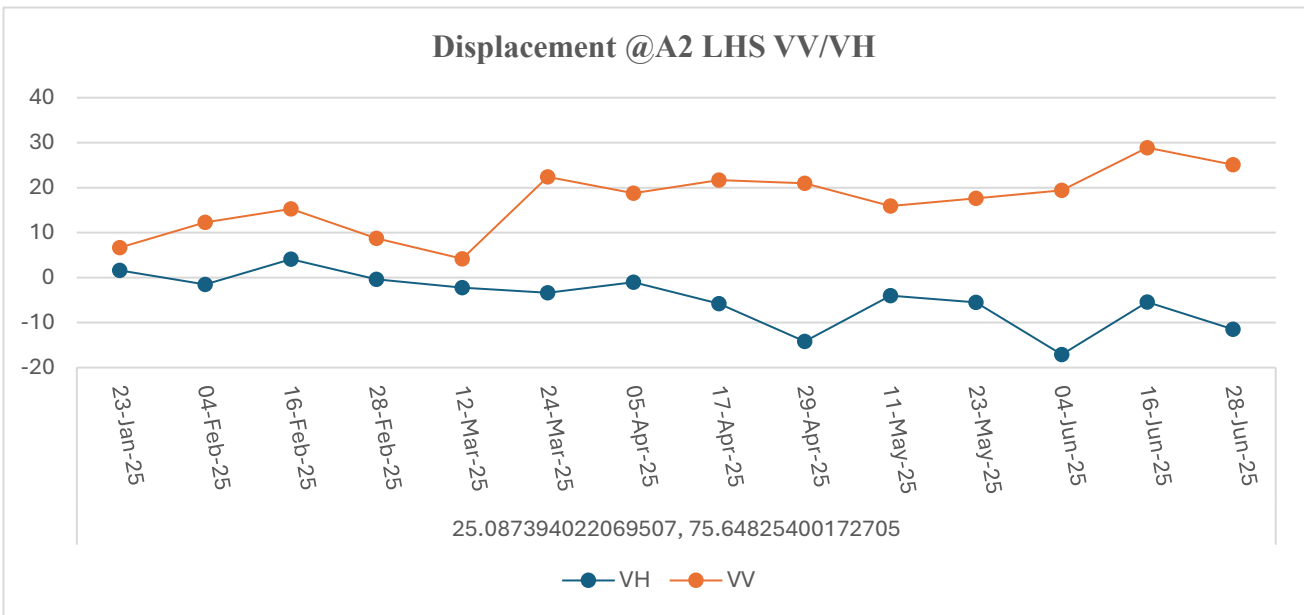
Critical Events & Anomalies:

- **January 2025:** VH begins at –8 mm, while VV is positive (+3 mm), indicating early lateral settlement occurring alongside mild vertical uplift.
- **June 2025:** VH recovers slightly to –4 mm before declining again to –11 mm, while VV falls to –3 mm, confirming continued deformation activity toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.087394022069507, 75.64825400172705)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability with predominantly negative values, ranging roughly between +4 mm and –15 mm. After small oscillations around zero in January–February.
- **Vertical Displacement (VV)** Vertical displacement remains consistently positive throughout the monitoring period (Jan–Jun 2025), indicating sustained vertical uplift at the A2 LHS location. VV values range approximately from +5 mm to +30 mm.

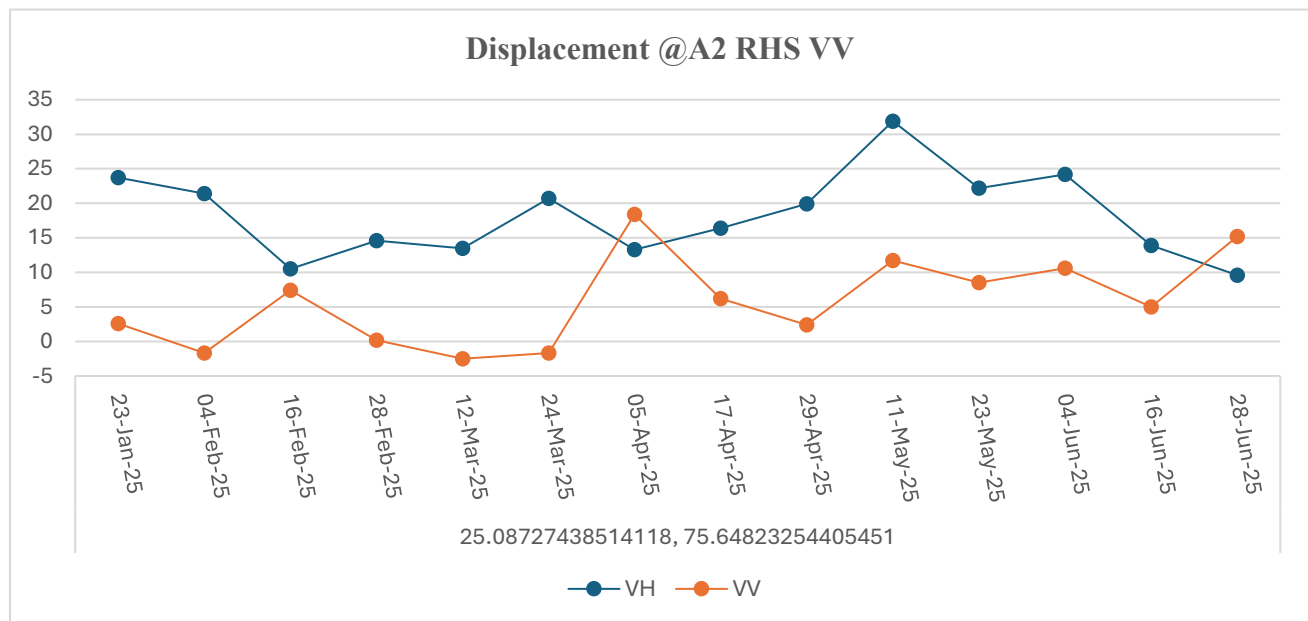
Critical Events & Anomalies:

- **February 2025:** VH briefly drops below 0 mm (–2 mm), while VV increases to +12 mm, indicating an early decoupling where vertical uplift occurs despite lateral settlement.
- **June 2025:** VV reaches its maximum values (+28–30 mm), while VH improves marginally to –6 to –8 mm, confirming continued uplift with unresolved lateral settlement toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.08727438514118, 75.64823254405451)



Key Observations:

Displacement Magnitude Patterns:

- Vertical (VV) displacement** Vertical displacement shows moderate variability with an overall positive tendency over the monitoring period (Jan–Jun 2025). VV values range approximately from –3 mm to +15 mm.
- Horizontal (VH) displacement** Horizontal displacement remains consistently positive with high variability, ranging roughly between +10 mm and +32 mm. VH exhibits multiple pronounced peaks, indicating strong and sustained lateral displacement throughout the monitoring period.

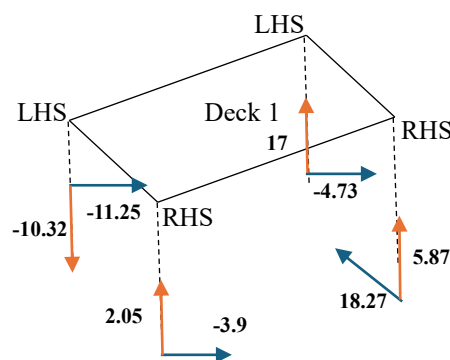
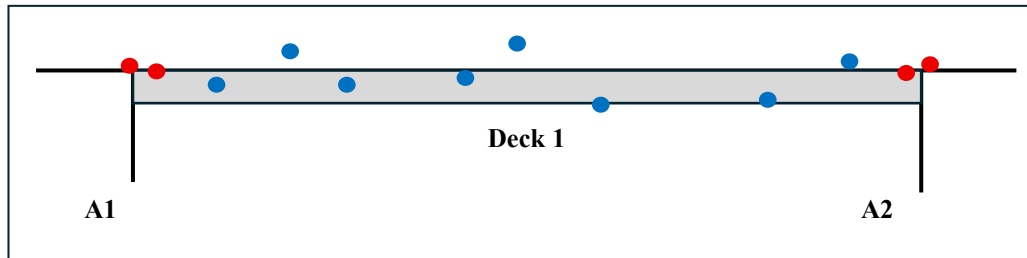
Critical Events & Anomalies:

- February 2025:** VH drops to +10 mm, while VV increases briefly to +7 mm, marking a temporary reduction in lateral displacement accompanied by vertical uplift.
- June 2025:** VH settles near +10 mm, while VV increases again to +15 mm, marking renewed vertical uplift toward the end of the monitoring period.

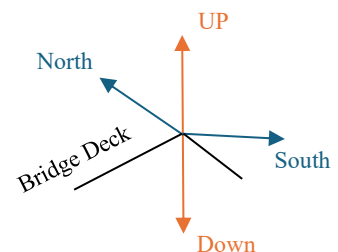
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly asymmetric vertical displacement, indicating significant differential vertical response between the LHS and RHS supports. The maximum downward settlement is observed at the LHS lower region, reaching approximately -10.32 mm, identifying this as the most critical vertical deformation zone.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates strong directional variability across the deck width, clearly reflecting torsional deformation behaviour. The maximum horizontal displacement occurs at the RHS lower region, reaching approximately $+18.27$ mm, representing dominant lateral translation on the RHS.

CONCLUSION:

- The bridge deck slab displays a torsion-dominated deformation response, with significant lateral displacement combined with asymmetric vertical movement. Overall, the deformation pattern suggests structurally significant behaviour, warranting continued monitoring and correlation with loading conditions and environmental effects.